BCSE429P

Cyber Physical Systems Design Lab

Laboratory 8 - Emission Models of Custom designed Junctions and Road Network

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Question:

Exercise 1 - Replicate the example as it is

Design a custom Road as given in the Video and the Blog Post, You may copy paste the source codes in a folder MyCustomRoad/

Play Video

SUMO Emission of Vehicles | VANETs

Exercise 2 - Design of New Road type

Design the following road and enable the emission model when emulating 100 vehicles on the road and plot their emissions like CO, CO2, NOx, HC and PM which is generated from the csv files.

(You can refer the above video and the blogpost as well)

PlainXML - SUMO Documentation (For Documentation of Node and Edge XML files)

1. Replicate the example as it is

Node.xml:

```
<nodes> <!-- The opening tag -->
  <node id="0" x="400.0" y="700.0" type="priority"/> <!-- def. of node "0" -->
  <node id="1" x="700.0" y="700.0" type="priority"/> <!-- def. of node "1" -->
  <node id="2" x="900.0" y="500.0" type="traffic_light"/>
  <node id="3" x="700.0" y="300.0" type="priority"/>
  <node id="4" x="400.0" y="300.0" type="priority"/> <!-- def. of node "4" -->
  <node id="5" x="200.0" y="500.0" type="traffic_light"/>
</nodes> <!-- The closing tag -->
```

Edge.xml:

```
<edges>
  <edge id="l01" from="0" to="1" priority="2" numLanes="2" speed="11.11"/>
  <edge id="r01" from="1" to="0" priority="3" numLanes="2" speed="13.89"/>

  <edge id="u12" from="1" to="2" priority="1" numLanes="2" speed="11.11"/>
  <edge id="d12" from="2" to="1" priority="2" numLanes="2" speed="11.11"/>

  <edge id="d23" from="2" to="3" priority="3" numLanes="2" speed="13.89"/>
  <edge id="u23" from="3" to="2" priority="1" numLanes="2" speed="11.11"/>

  <edge id="l43" from="3" to="4" priority="2" numLanes="2" speed="11.11"/>
  <edge id="r43" from="4" to="3" priority="3" numLanes="2" speed="13.89"/>

  <edge id="d54" from="5" to="4" priority="1" numLanes="2" speed="11.11"/>
  <edge id="u54" from="4" to="5" priority="2" numLanes="2" speed="11.11"/>

  <edge id="d05" from="0" to="5" priority="3" numLanes="2" speed="13.89"/>
  <edge id="u05" from="5" to="0" priority="1" numLanes="2" speed="11.11"/>

  <edge id="l52" from="5" to="2" priority="3" numLanes="2" speed="13.89"/>
  <edge id="r52" from="2" to="5" priority="1" numLanes="2" speed="11.11"/>

</edges>
```

Config:

```
<configuration>
<input>
<net-file value="file.net.xml"/>
<route-files value="file.rou.xml"/>
```

```
</input>
<time>
<begin value="0"/>
<end value="1000"/>
<step-length value="0.5"/>
</time>
</configuration>
```

Commands:

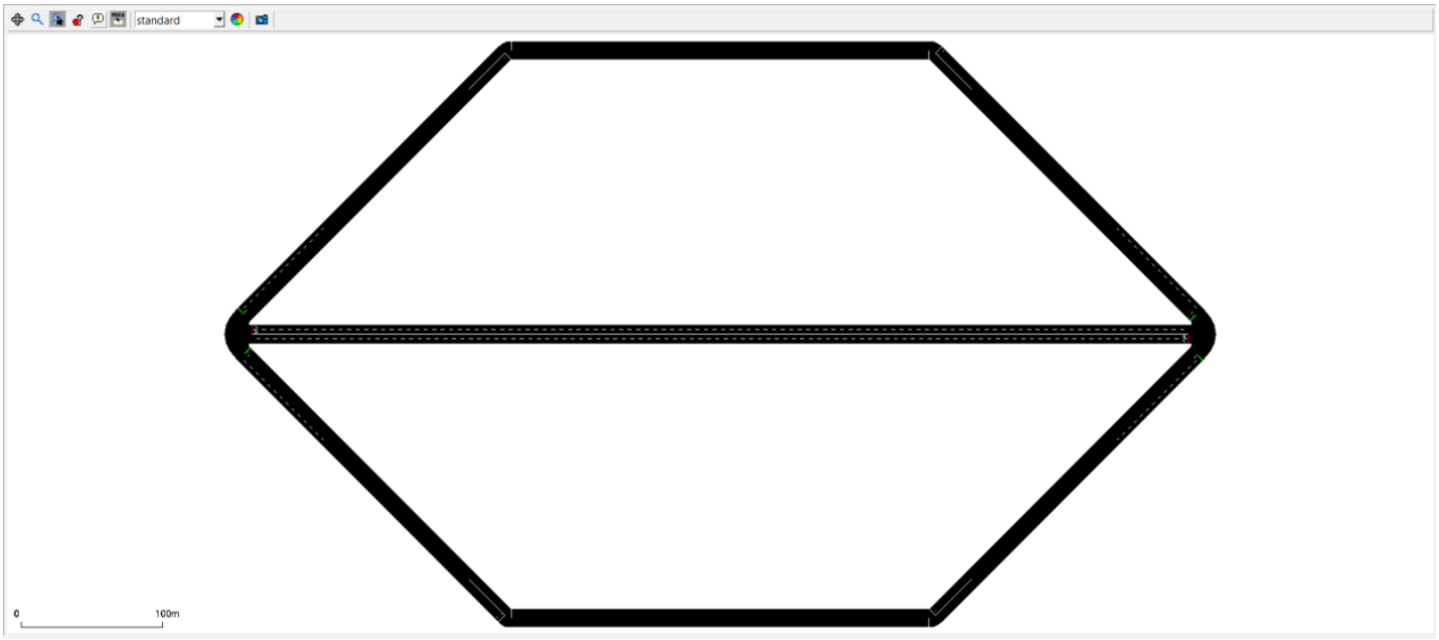
1) Obtaining the network:
netconvert -n file.nod.xml -e file.edg.xml -o file.net.xml

2) Opening the created network
Sumo-gui file.sumo.cfg

3) Emission and other outputs:
sumo -c file.sumo.cfg –emission-output emission.xml – battery-output battery.xml –fcd-ouput car.xml – summary-output summary.xml –collision-output collision.xml

4) Convert xml to csv:
python sumo/tools/xml/xml2csv.py emission.xml

Output:



2. Design of New Road type

Node.xml:

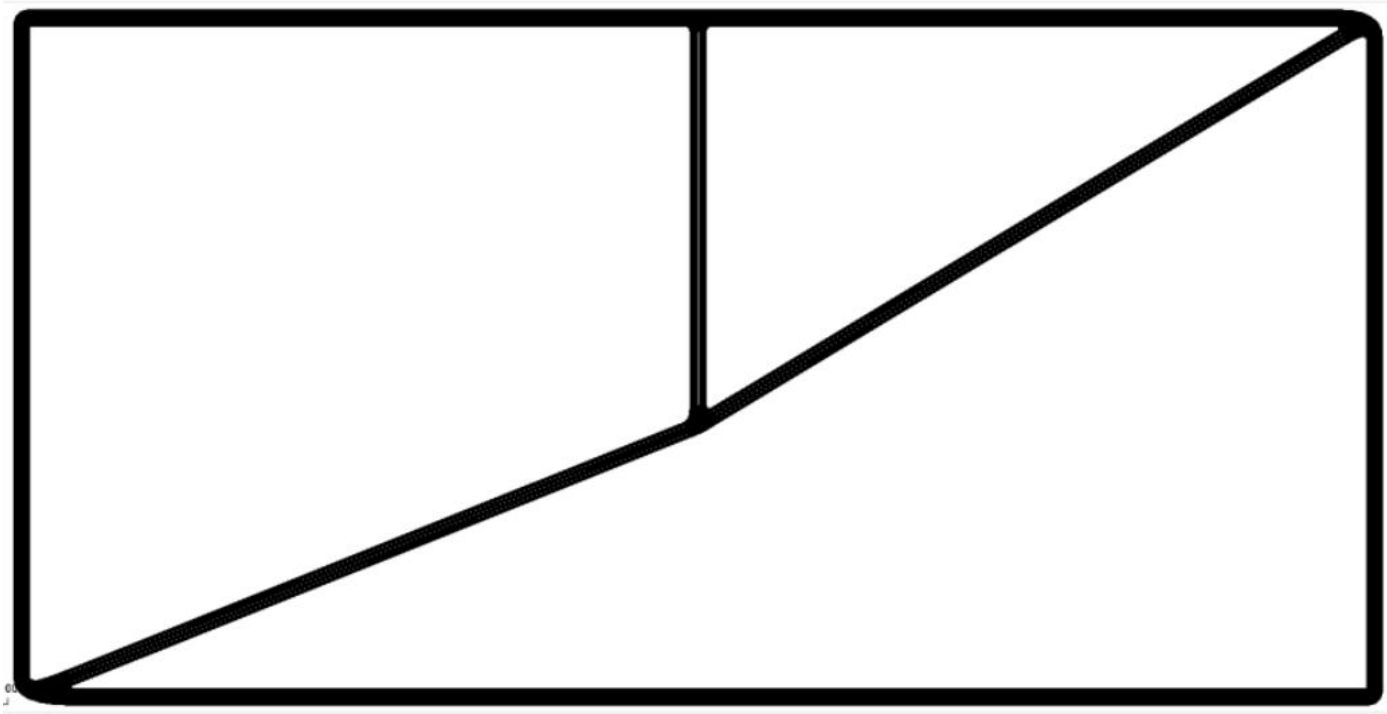
```
<nodes>
    <node id="0" x="0.0" y="500.0" type="priority"/>
    <node id="1" x="500.0" y="500.0" type="priority"/>
    <node id="2" x="1000.0" y="500.0" type="traffic_light"/>
    <node id="3" x="500.0" y="200.0" type="traffic_light"/>
    <node id="4" x="0.0" y="0.0" type="priority"/>
    <node id="5" x="1000.0" y="0.0" type="priority"/>
</nodes>
```

Edge.xml:

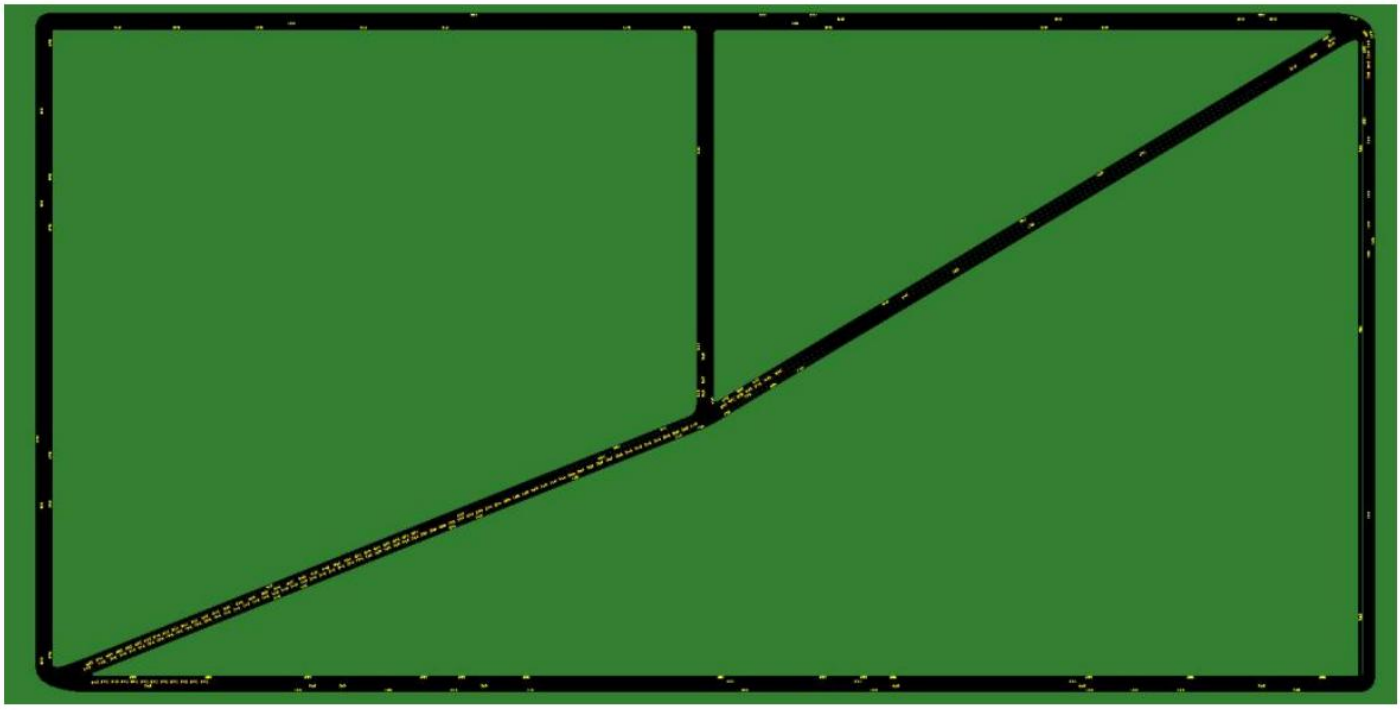
```
<edges>
    <!-- Top horizontal road -->
    <edge id="e_0_1" from="0" to="1" numLanes="2" speed="13.9"/>
    <edge id="e_1_0" from="1" to="0" numLanes="2" speed="13.9"/>
    <edge id="e_1_2" from="1" to="2" numLanes="2" speed="13.9"/>
    <edge id="e_2_1" from="2" to="1" numLanes="2" speed="13.9"/>
    <!-- Left vertical road -->
    <edge id="e_0_4" from="0" to="4" numLanes="2" speed="13.9"/>
    <edge id="e_4_0" from="4" to="0" numLanes="2" speed="13.9"/>
    <!-- Inner connections through node 3 -->
    <edge id="e_1_3" from="1" to="3" numLanes="2" speed="13.9"/>
    <edge id="e_3_1" from="3" to="1" numLanes="2" speed="13.9"/>
    <edge id="e_2_3" from="2" to="3" numLanes="2" speed="13.9"/>
    <edge id="e_3_2" from="3" to="2" numLanes="2" speed="13.9"/>
    <edge id="e_4_3" from="4" to="3" numLanes="2" speed="13.9"/>
    <edge id="e_3_4" from="3" to="4" numLanes="2" speed="13.9"/>
    <!-- Right vertical road -->
    <edge id="e_2_5" from="2" to="5" numLanes="2" speed="13.9"/>
    <edge id="e_5_2" from="5" to="2" numLanes="2" speed="13.9"/>
    <!-- Bottom horizontal road -->
    <edge id="e_4_5" from="4" to="5" numLanes="2" speed="13.9"/>
    <edge id="e_5_4" from="5" to="4" numLanes="2" speed="13.9"/>
</edges>
```

Output:

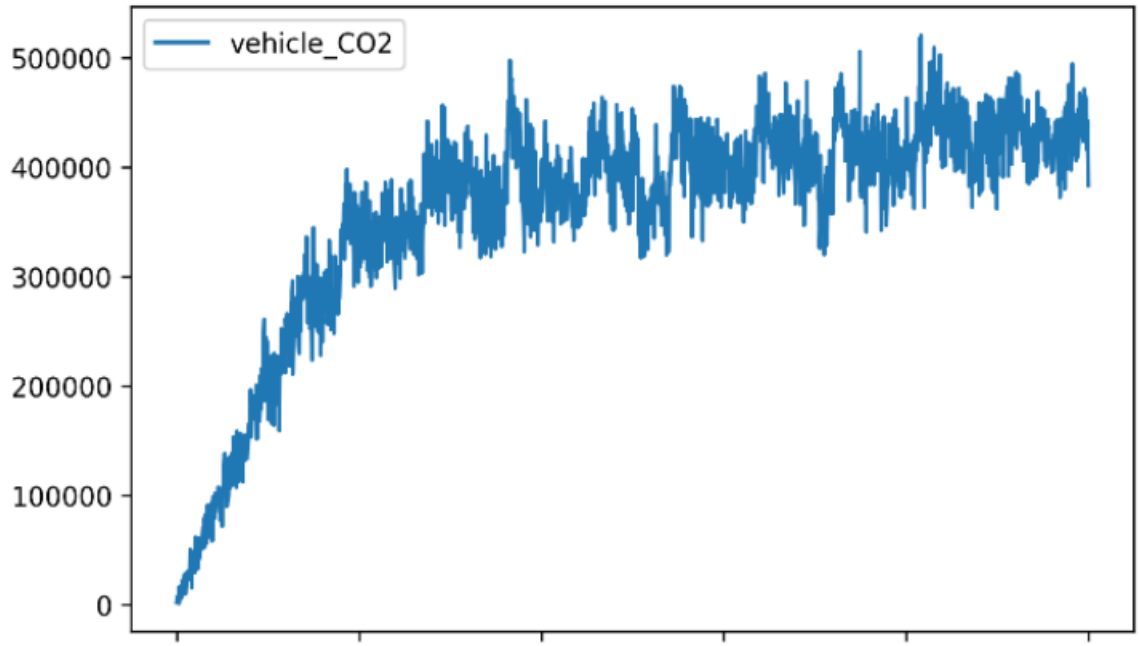
Network created

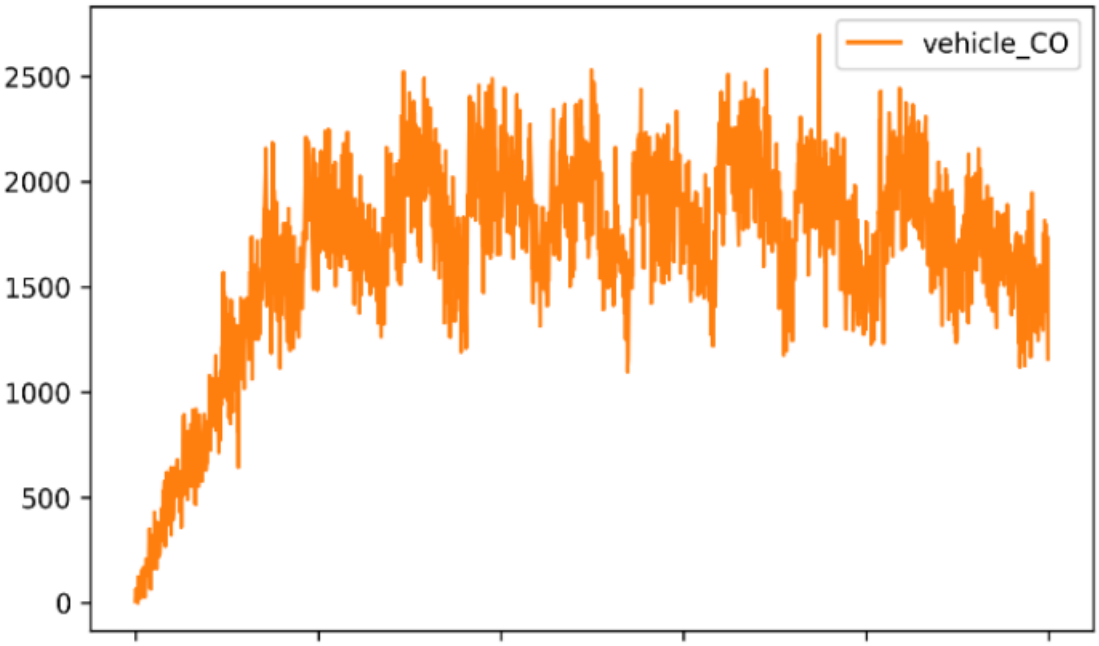


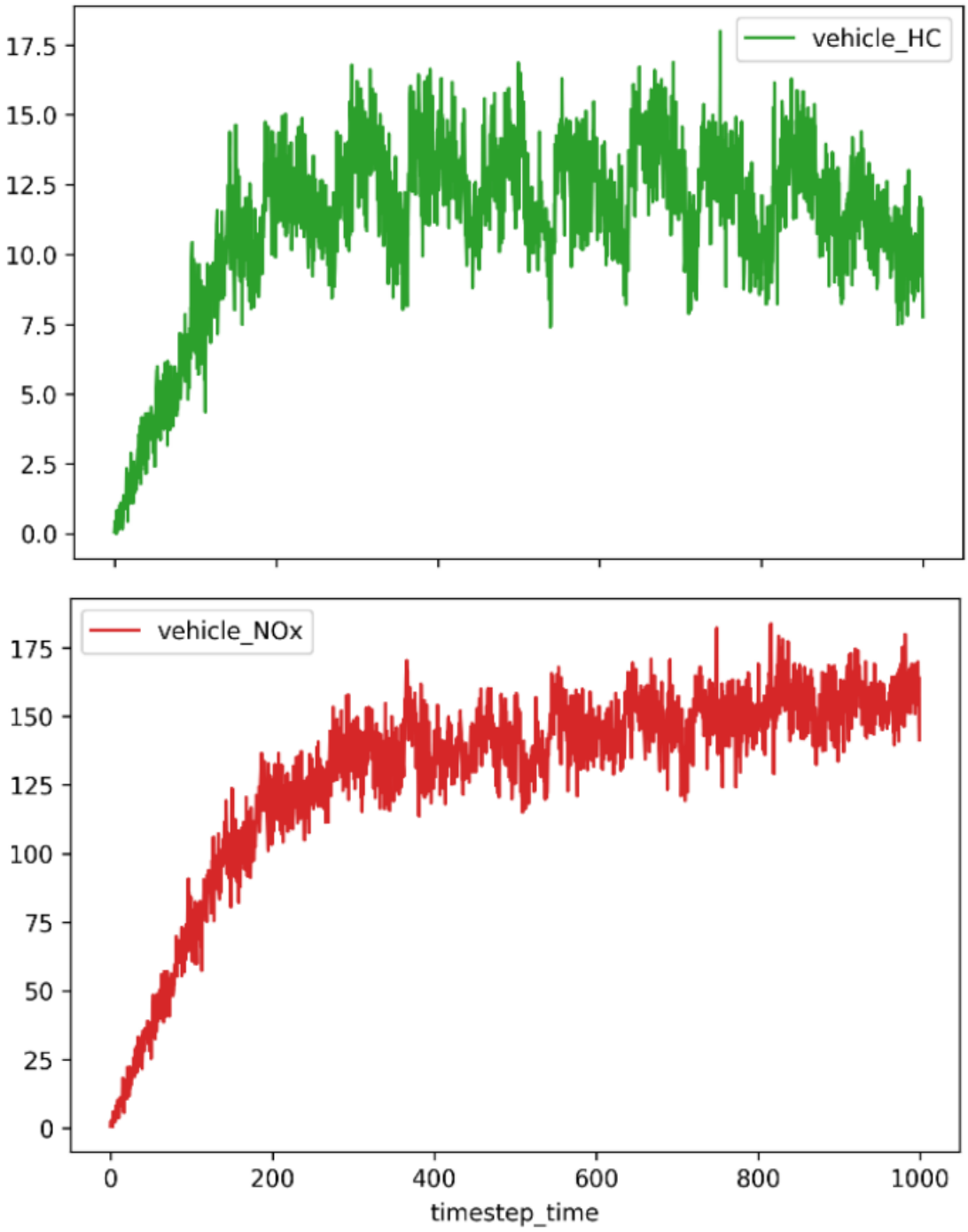
Simulation:

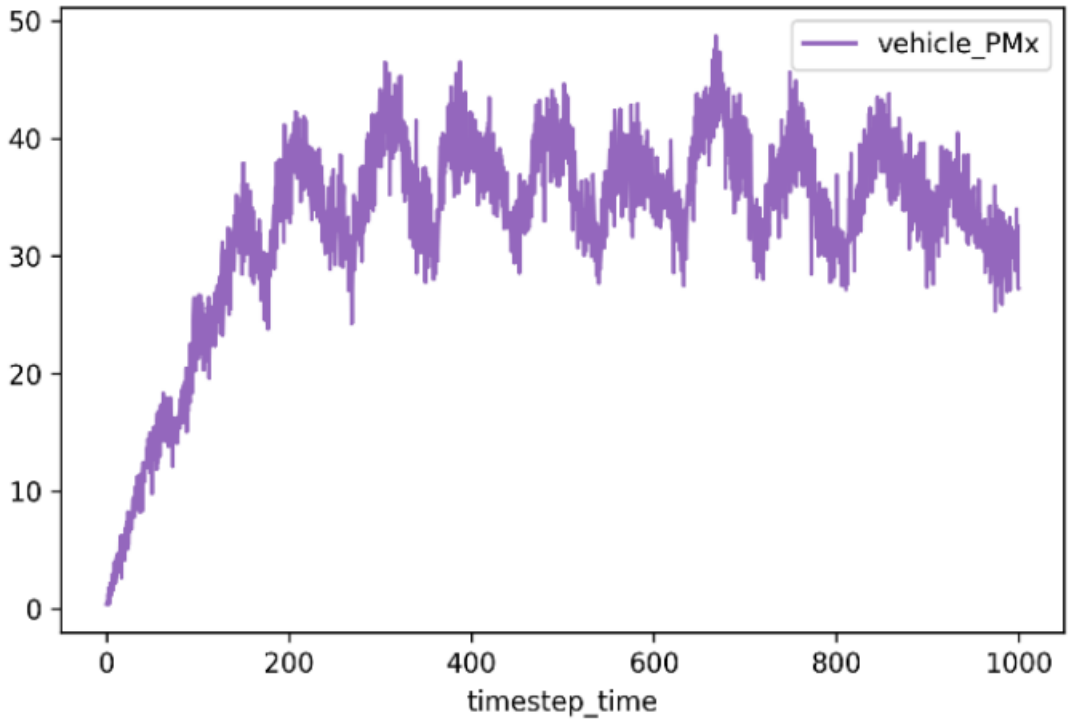


Plot of the emissions:






vehicle_HC
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
vehicle_NOx
175
150
125
100
75
50
25
0
0
200
400
600
800
1000
timestep_time


vehicle_PMx
50
40
30
20
10
0
0
200
400
600
800
1000
timestep_time